EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**RLE always reduces the size of any input.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

## Lesson 128 · Run-Length Encoding: Encoding and Decod

Name: _____ Class: _____ Date: _____

**1. When must lossless compression be used?**

**2. In a queue, what does FIFO stand for?**

**3. What is a function parameter?**

**4. In a BST, which side holds a smaller value?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

When exact reconstruction is necessary, such as executable code.

**2. Model answer**

First in, first out.

**3. Model answer**

An input used by a subroutine.

**4. Model answer**

The left subtree.

**By the end, you can…**

- Group identical adjacent symbols, not identical symbols separated by others.
- Record every run as count and symbol consistently.
- Count storage using an explicit format, including counts and delimiters.

**Vocabulary — know these terms**

RLE — Encodes consecutive equal symbols as a symbol and count.

run — A consecutive sequence of identical symbols.

overhead — Extra data required by an encoding scheme.

lossless — Decoding reproduces the original exactly.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

A bitmap row has 40 alternating pixels. Explain why RLE might increase file size and identify a better-suited pattern. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

Apply RLE to AAAABBBCCAAAA using count-symbol pairs. [6]

**Question 2**

Decode (3,X)(1,Y)(2,X) and explain why runs must retain order. [5]





NEW LEARNING

# Read, annotate and explain

**Identify runs**

Group identical adjacent symbols, not identical symbols separated by others.

**Encode**

Record every run as count and symbol consistently.

**Check cost**

Count storage using an explicit format, including counts and delimiters.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Encode AAAAABBCCC using (count,symbol) pairs; verify the result.**

**AAAAA | BB | CCC**

Predict the first step from the prompt.

**(5,A)(2,B)(3,C)**

Explain the intermediate result or decision.

**Decode original**

AAAAA =5A, BB=2B, CCC=3C. Encoded result (5,A)(2,B)(3,C); decoded data restores AAAAABBCCC exactly.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve